Rendered Bayesian Belief Network MCMC

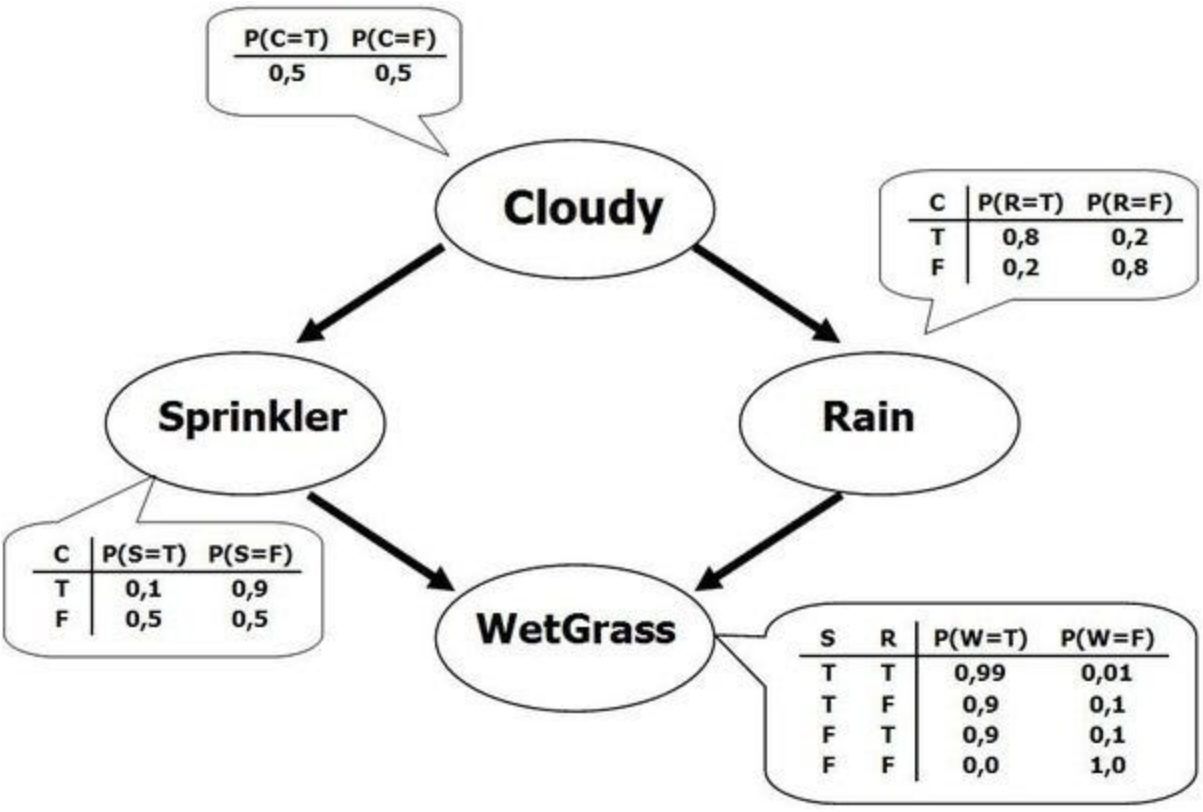
I presented this notebook to teach graduate and postgraduate scientists about probabilistic coding in 2020. I chose this example to submit in support of my Astra Residency Fellowship application because:

1. It shows how to use Monte Carlo Markov Chain (MCMC) sampling in python; MCMC sampling is useful because:
   - MCMC sampling gives an unbiased estimate of the posterior, in contrast to Variational Inference which is often biased
   - It is robust, in that it gives both uncertainty bounds for parameter estimates, and diagnostics for debugging the inference pipeline.
   - The sampler can be hardware accelerated using GPUs to converge on a solution
2. It uses Bayesian Belief Networks, which formalize both using and inferring priors from data
3. It shows how to use probabilistic programming within the python ecosystem
4. It's short enough to be understandable

The original presentation was video taped as part of the NASA ICESat-2 Hackweek, and can be found here. Other good examples of my programming skills can be seen in the Sklearn implementation of the OPTICS clustering algorithm, which I am the primary author of (original pull request, in excess of 1,000 lines of code), or a shorter merged pull request to speed up multivariate normal sampling for the CuPy library.

Inference using Bayesian Networks

Inference is a powerful technique that combines empirical evidence and prior statistical beliefs to determine likelihood from causal connections. A classic example is presented below in graphical form (taken from this blog; please excuse the convention of substituting commas for decimal places):



As shown above, there are two causal explanations for why the grass is wet-- the sprinkler could be on, or it could have rained. We can enter the graph at any point to infer the conditional likelihood of each variable; for example, if we know the state of *cloudy*, we can replace the 50/50 prior probability, and propagate that through the graph. We can also reverse the direction; if we know that the grass is wet, we can infer the empirical likelihood of *cloudy* being true such that it is weighted by this evidence, instead of just the prior belief. Note that this means that it is possible to determine prior beliefs from data, by choosing an naive prior and replacing after acquiring empirical data.

Numeric Considerations

Bayesian nets can be used as simple feed forward networks (if we are confident in our priors), or trained to determine the conditional probabilities that the network itself uses. We'll use pymc3 in this notebook to examine how to formulate and build Bayesian networks-- first using discrete variables, then expanding to incorporate continuous variables. All of this falls under ***probabilistic programming***; there a wide variety of tasks that the framework can be used for. The pymc3 library uses Monte Carlo simulations for inference, which can sped up substantially by GPUs-- although they run just fine on CPUs as well.

Defining a Probabilistic Model

We'll use a worked epidemiology example to show how to set both static parameters, and distributions over parameters, and then explore adding in empirical observations to the model.

In [4]:
```
Image('./a4Murphy1.jpg')
```

Out[4]:

Consider the DGM in Figure 20.10 which represents the following fictitious biological model. Each G_i represents the genotype of a person: $G_i = 1$ if they have a healthy gene and $G_i = 2$ if they have an unhealthy gene. G_2 and G_3 may inherit the unhealthy gene from their parent G_1. $X_i \in \mathbb{R}$ is a continuous measure of blood pressure, which is low if you are healthy and high if you are unhealthy. We define the CPDs as follows

$$p(G_1) = [0.5, 0.5] \tag{20.73}$$

$$p(G_2|G_1) = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix} \tag{20.74}$$

$$p(G_3|G_1) = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix} \tag{20.75}$$

$$p(X_i|G_i = 1) = \mathcal{N}(X_i|\mu = 50, \sigma^2 = 10) \tag{20.76}$$

$$p(X_i|G_i = 2) = \mathcal{N}(X_i|\mu = 60, \sigma^2 = 10) \tag{20.77}$$

The meaning of the matrix for $p(G_2|G_1)$ is that $p(G_2 = 1|G_1 = 1) = 0.9$, $p(G_2 = 1|G_1 = 2) = 0.1$, etc.

In [1]:
```
%pylab inline
```

```
%pylab is deprecated, use %matplotlib inline and import the required libraries.
Populating the interactive namespace from numpy and matplotlib
```

In [2]:
```
from IPython.display import Image
from IPython.display import Markdown as md
```

In [3]:
```
import pymc3 as pm
```

```
from pymc3 import Normal, Model
from pymc3.math import switch
```

In [5]:
```
# General model
Image('./a4Murphy2.jpg')
```

Out[5]:

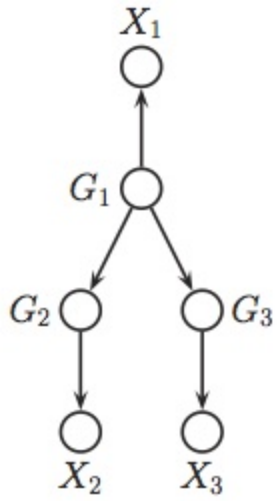
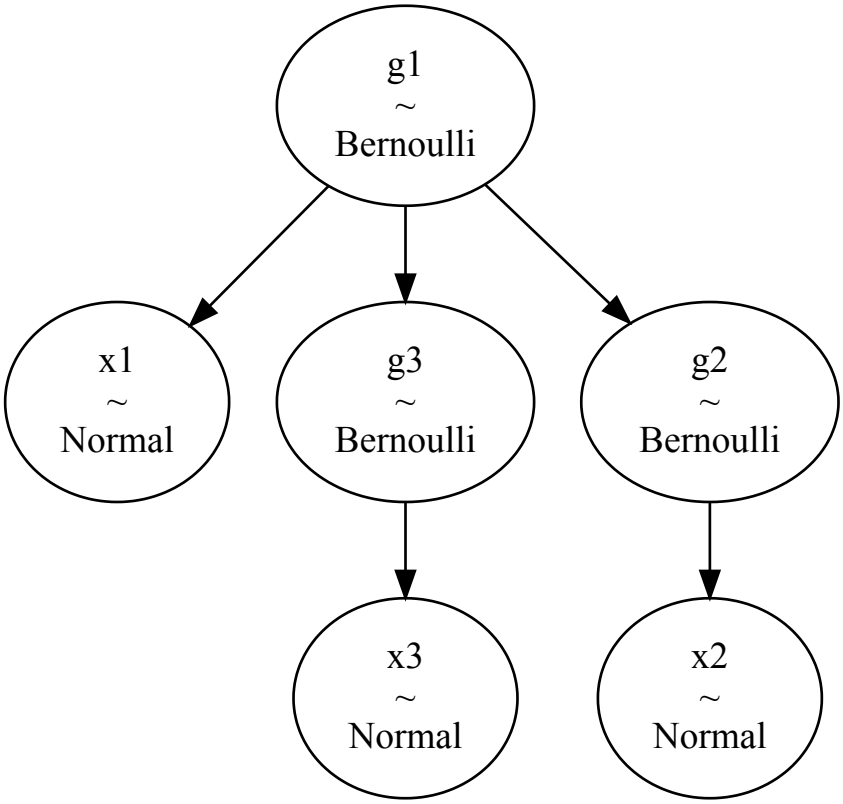


Figure 20.10 A simple DAG representing inherited diseases.

In [6]:
```
with Model() as bayesnet:
    # Parent
    g1_healthy = pm.distributions.discrete.Bernoulli('g1',0.5)

    # Latent variable that conditions the children
    g1_L = switch(g1_healthy > 0.5, 0.9, 0.1)

    # Left Child
    g2_healthy = pm.distributions.discrete.Bernoulli('g2', g1_L)
    # Right Child
    g3_healthy = pm.distributions.discrete.Bernoulli('g3', g1_L)
    # Conditional switches for blood pressure mean
    rate1 = switch(g1_healthy > .5, 50, 60)
    rate2 = switch(g2_healthy > .5, 50, 60)
    rate3 = switch(g3_healthy > .5, 50, 60)
    # Parent and children
    x1 = Normal('x1', mu=rate1, sd=10**.5)
    x2 = Normal('x2', mu=rate2, sd=10**.5)
    x3 = Normal('x3', mu=rate3, sd=10**.5)
```

In [7]:
```
pm.model_graph.model_to_graphviz(model=bayesnet)
```

Out[7]:



We see that the defined model is identical to the reference figure, just with X_1 being placed in below rather than above. The *'G'* parameters are all Bernoulli distributions that specify a binary outcome. The *'$X_$'** variables are continuous distributions of blood pressure.

Exercise 1

Compute P(G1 = 2 | X2 = 50). This is the probability that the parent (G_1) has the heredity disease given that the child (X_2) has blood pressure of 50.

In [8]:

```
# Compute P(G1 = 2 | X2 = 50)

with Model() as bayesnet:
    # Parent
    g1_healthy = pm.distributions.discrete.Bernoulli('g1',0.5)

    # Latent variable that conditions the children
    g1_L = switch(g1_healthy > 0.5, 0.9, 0.1)

    # Left Child
    g2_healthy = pm.distributions.discrete.Bernoulli('g2', g1_L)
    # Right Child
    g3_healthy = pm.distributions.discrete.Bernoulli('g3', g1_L)
    # Conditional switches for blood pressure mean
    rate1 = switch(g1_healthy > .5, 50, 60)
    rate2 = switch(g2_healthy > .5, 50, 60)
    rate3 = switch(g3_healthy > .5, 50, 60)
    # Grandkids and child
    x1 = Normal('x1', mu=rate1, sd=10**.5)
    # Setting x2 as an observed variable
    x2 = Normal('x2', mu=rate2, sd=10**.5, observed=50)
    x3 = Normal('x3', mu=rate3, sd=10**.5)
```

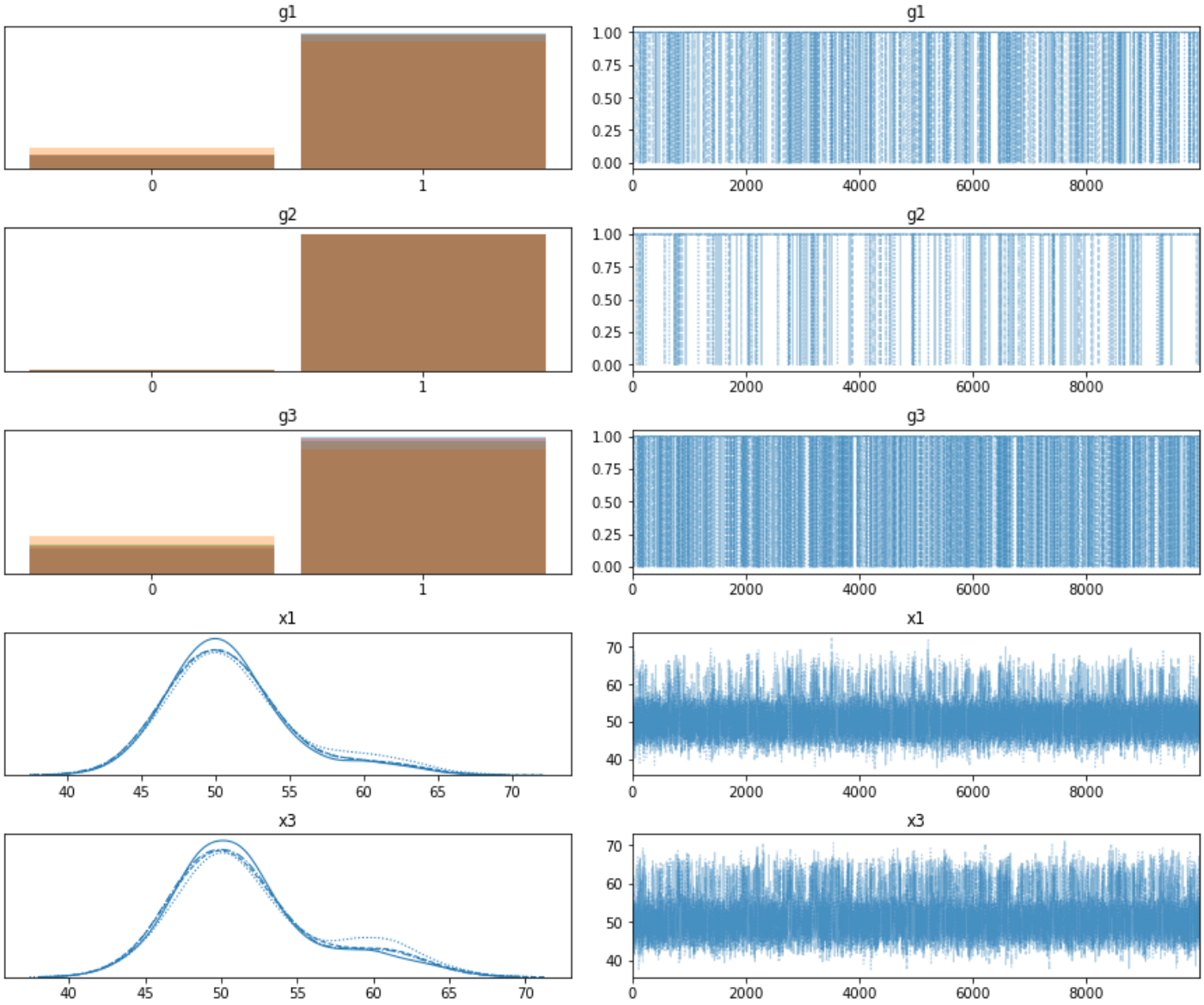
Note that the above is identical to our original model definition-- the only thing that has changed, is that we have set X_2 as an observed variable. There is an analytic solution to this type of inference, but we'll use sampling to estimate an empirical solution:

In [9]:
```
with bayesnet:
    mytrace = pm.sample(10000, tune=2000)
```

```
Multiprocess sampling (4 chains in 4 jobs)
CompoundStep
>BinaryGibbsMetropolis: [g1, g2, g3]
>NUTS: [x3, x1]
Sampling 4 chains, 0 divergences: 100%|██████████| 48000/48000 [04:05<00:00, 195.13draws/
s]
The number of effective samples is smaller than 10% for some parameters.
```

We see that PyMC3 does the hard work of choosing the appropriate samplers for us :-) The binary variables use Gibbs Metropolis-Hastings, and the continuous use the No-U-Turn-Sampler. The variable X_2 is not sampled, since we set it as observed. The result of the sampling is a *trace*; we can view the trace to get our answer to the original question (although it will take a second to plot given the sample size):

In [10]:
```
pm.traceplot(mytrace);
```



The original question was "*what's the probability that the parent has the disease*", so we can take the number of samples where $G_1 = 1$ (is healthy), and divide by the total number of samples to get the percentage of the population in the simulation that was healthy:

In [11]:
```
p_g1_healthy = sum(mytrace['g1'])/len(mytrace['g1'])
```

```
p_g1_not_healthy = 1 - p_g1_healthy
p_g1_healthy, p_g1_not_healthy
```

Out[11]:
```
(0.893325, 0.10667499999999996)
```

In [12]:
```
md("""We find {}% of the G1 samples are healthy (G1 = 1); so  {}% are unhealthy (G1 = 2).
   We can actually access the same information using the `summary()` function, which will
   provide an error estimate:""".format(p_g1_healthy.round(4)*100,p_g1_not_healthy.round(4
```

Out[12]: We find 89.33% of the G1 samples are healthy (G1 = 1); so 10.67% are unhealthy (G1 = 2). We can actually access the same information using the `summary()` function, which will also provide an error estimate:

In [13]:
```
pm.summary(mytrace, round_to=5)
```

Out[13]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_mean	ess_sd	ess_bulk
g1	0.89332	0.30870	0.00000	1.00000	0.00761	0.00538	1647.12675	1647.12675	1647.12675
g2	0.99218	0.08811	1.00000	1.00000	0.00069	0.00049	16090.00603	16090.00603	16090.00603
g3	0.81540	0.38798	0.00000	1.00000	0.00903	0.00639	1846.68906	1846.68906	1846.68906
x1	51.09243	4.44766	43.55519	60.89222	0.08253	0.06028	2904.40952	2722.81419	4115.48051
x3	51.85390	5.00411	43.97496	62.68362	0.09775	0.07059	2620.91546	2513.53192	3458.31141

In [14]:
```
stats = pm.summary(mytrace, 'g1', round_to=5)
err = round(float(stats.mcse_mean),5)
plus = round(float(1-stats['mean'] + stats.mcse_mean),5)
minus = round(float(1-stats['mean'] - stats.mcse_mean),5)
ans = round(float(1-stats['mean']),5)

md("""As mentioned, we could analytically compute the answer:

- The analytic answer is 0.10535
- The MCMC estimate is {} +/- {}

Since 0.10535 is between {} and {}, seems that we did well.""".format(ans,err,plus,minus))
```

Out[14]: As mentioned, we could analytically compute the answer:

- The analytic answer is 0.10535
- The MCMC estimate is 0.10668 +/- 0.00761

Since 0.10535 is between 0.11429 and 0.09907, seems that we did well.

Brief digression on MCMC error

I was curious about the MCMC error... does the sampler actually know how good a job it's doing? Running the sampler again with a factor less samples certainly gives a worse estimate:

In [15]:
```
with bayesnet:
    mytrace = pm.sample(2000);
```

```
Multiprocess sampling (4 chains in 4 jobs)
CompoundStep
>BinaryGibbsMetropolis: [g1, g2, g3]
```

```
>NUTS: [x3, x1]
Sampling 4 chains, 0 divergences: 100%|██████████| 10000/10000 [00:24<00:00, 400.49draws/
s]
The number of effective samples is smaller than 10% for some parameters.
```

In [16]:
```
p_g1_healthy = sum(mytrace['g1'])/len(mytrace['g1'])
p_g1_not_healthy = 1 - p_g1_healthy
p_g1_healthy, p_g1_not_healthy
```

Out[16]:
```
(0.880625, 0.11937500000000001)
```

In [17]:
```
pm.summary(mytrace, round_to=5)
```

Out[17]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_mean	ess_sd	ess_bulk	
g1	0.88062	0.32425	0.00000	1.00000	0.01734	0.01227	349.79473	349.79473	349.79473	34
g2	0.99450	0.07396	1.00000	1.00000	0.00094	0.00067	6144.51864	6144.51864	6144.51864	800
g3	0.78875	0.40822	0.00000	1.00000	0.02125	0.01504	369.11995	369.11995	369.11995	36
x1	51.16394	4.47708	43.76447	61.16656	0.18439	0.13429	589.53421	556.29684	809.87079	77
x3	52.13296	5.13962	44.55922	63.11791	0.22530	0.16210	520.40437	503.25999	687.08104	192

In [18]:
```
stats = pm.summary(mytrace, 'g1', round_to=5)
err = round(float(stats.mcse_mean),5)
plus = round(float(1-stats['mean'] + stats.mcse_mean),5)
minus = round(float(1-stats['mean'] - stats.mcse_mean),5)
ans = round(float(1-stats['mean']),5)

md("""The new answer is {} +/- {}. Given that the analytic answer is 0.10535, this
   is clearly a worse estimate... but 0.10535 still falls within the {} to {} bounded
   error envelope, so we know the answer is less precise. If we need more precision, we ca
   simply run more samples till the MC error is acceptably low.""".format(ans,err,minus,pl
```

Out[18]: The new answer is 0.11938 +/- 0.01734. Given that the analytic answer is 0.10535, this is clearly a worse estimate... but 0.10535 still falls within the 0.10204 to 0.13672 bounded error envelope, so we know the answer is less precise. If we need more precision, we can simply run more samples till the MC error is acceptably low.

Exercise 2

Compute P(X3 = 50 | X2 = 50). That is, what is the conditional probability that child 2 (X_3) has blood pressure equal to 50 given that their sibling (child 1, X_2) has normal blood pressure (50). As before, we simply fix the variable of interest using the observed flag:

In [19]:
```
with Model() as bayesnet:
    # Parent
    g1_healthy = pm.distributions.discrete.Bernoulli('g1',0.5)

    # Latent variable that conditions the children
    g1_L = switch(g1_healthy > 0.5, 0.9, 0.1)

    # Left Child
    g2_healthy = pm.distributions.discrete.Bernoulli('g2', g1_L)
    # Right Child
    g3_healthy = pm.distributions.discrete.Bernoulli('g3', g1_L)
```

```
    # Conditional switches for blood pressure mean
    rate1 = switch(g1_healthy > .5, 50, 60)
    rate2 = switch(g2_healthy > .5, 50, 60)
    rate3 = switch(g3_healthy > .5, 50, 60)
    # Grandkids and child
    x1 = Normal('x1', mu=rate1, sd=10**.5)
    x2 = Normal('x2', mu=rate2, sd=10**.5)
    # Setting x3 as an observed variable
    x3 = Normal('x3', mu=rate3, sd=10**.5, observed=50)
```

In [20]:

```
with bayesnet:
    mytrace = pm.sample(25000)
```

```
Multiprocess sampling (4 chains in 4 jobs)
CompoundStep
>BinaryGibbsMetropolis: [g1, g2, g3]
>NUTS: [x2, x1]
Sampling 4 chains, 0 divergences: 100%|██████████| 102000/102000 [04:21<00:00, 390.00draw
s/s]
The acceptance probability does not match the target. It is 0.8879690375868736, but should
be close to 0.8. Try to increase the number of tuning steps.
The number of effective samples is smaller than 10% for some parameters.
```

Because the variable that we're interested in is continuous, it isn't as obvious how to answer our question... we could of course cheat, and cast X_2 as discrete value:

In [21]:

```
n1 = mytrace['x2']>51
n2 = mytrace['x2']<50

(len(mytrace['x2']) - sum(n1) - sum(n2))/len(mytrace['x2'])
```

Out[21]:

```
0.09736
```

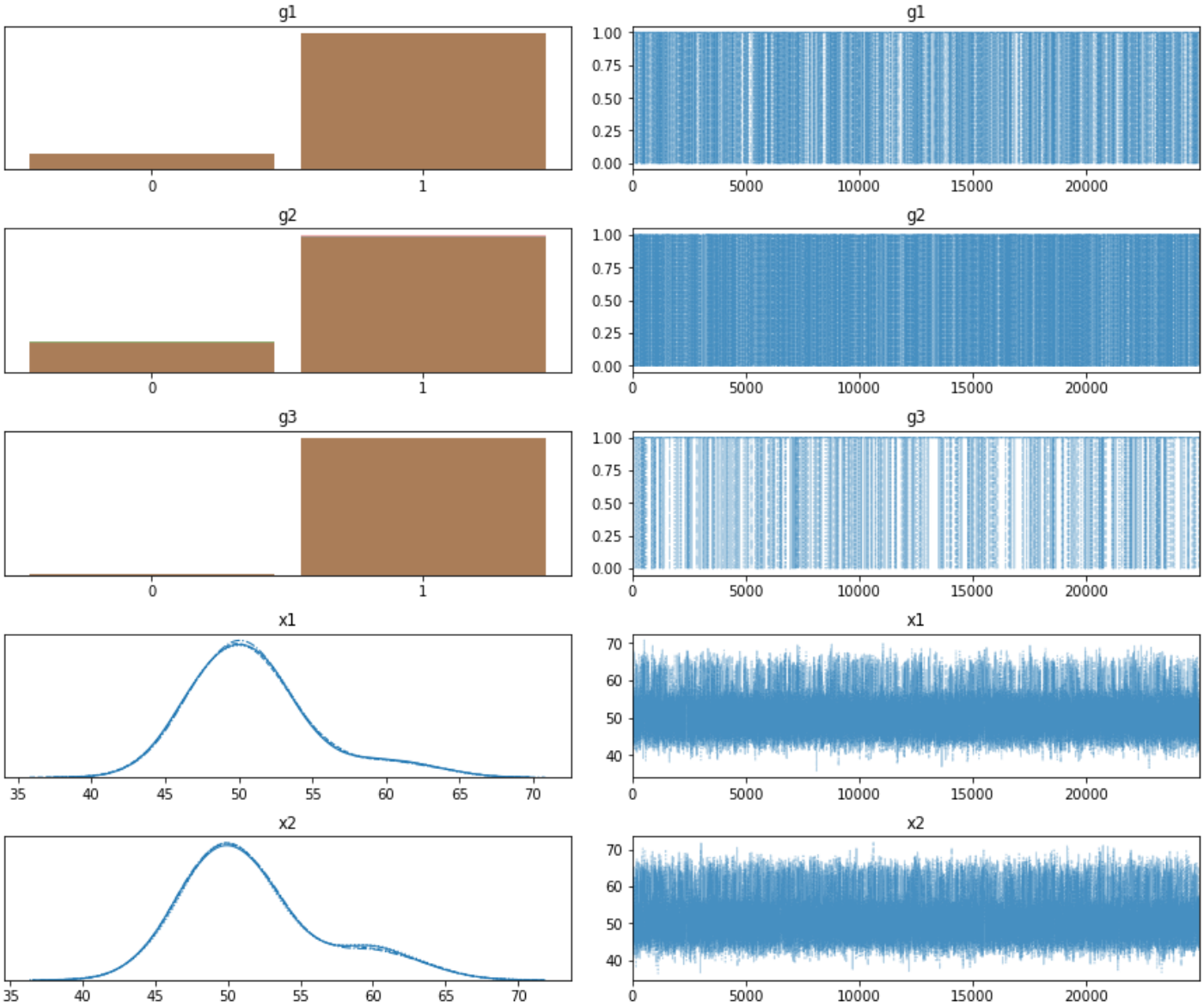
The analytic solution is 0.10306, so casting X_2 as a discrete variable gives a close approximation, but is also kinda unsatisfying because:

- The width for the approximation isn't always obvious. Here, for blood pressure, using a natural number is a pretty clear fit-- but this isn't as obvious for other cases.
- We don't have an error estimation on the approximation. This seems not very Bayesian.

We could also try to cheat an answer by reading X_2 directly off of the trace plots:

In [22]:

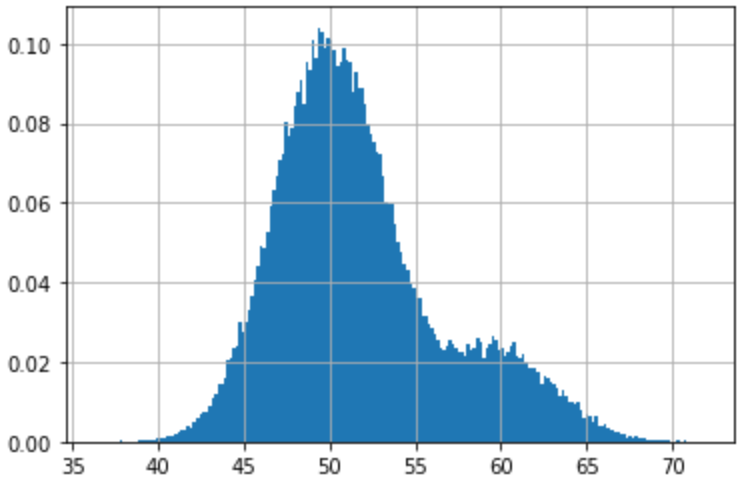
```
pm.traceplot(mytrace);
```



In [23]:
```
df = pm.trace_to_dataframe(mytrace)
```

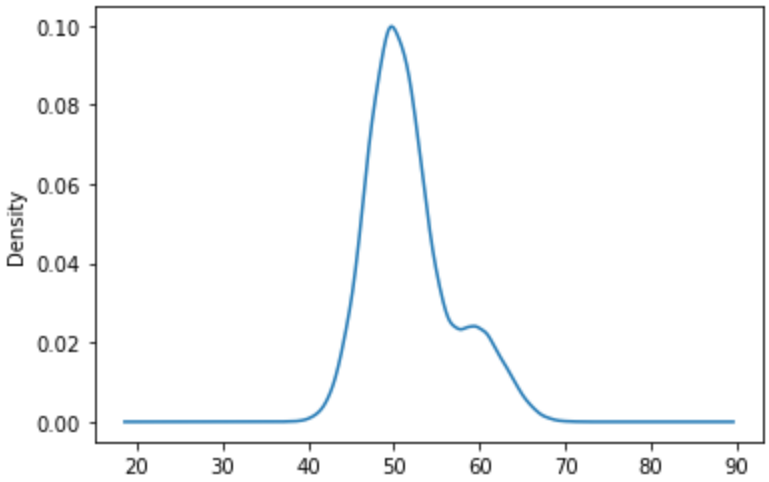
In [24]:
```
df.x2.hist(bins=200,density=True)
```

Out[24]:
```
<matplotlib.axes._subplots.AxesSubplot at 0x7f4274d0a850>
```



In [25]:
```
df.x2.plot.kde()
```

Out[25]: <matplotlib.axes._subplots.AxesSubplot at 0x7f427cb84b10>



Since we've fixed X_3, we don't see it in the plots, but we do get the X_2 variable that was previously omitted when it was fixed. We can see that the probability for X_2= 50 is close to our analytic solution, and we don't have to decide a width, but:

- We still don't have an error estimate on the parameter
- We're just eyeballing a value off of a graph. There's surely a more numerically precise way to do this...

Estimating a numerically robust solution

My preferred solution is to estimate an empirical PDF (Probability Density Function) via KDE (Kernel Density Estimation) using the trace of X_2 as the input. This allows probability estimation of any value, including decimal values:

In [26]:
```
from scipy.stats.kde import gaussian_kde
```

In [27]:
```
my_pdf = gaussian_kde(mytrace['x2'])
```

In [28]:
```
my_pdf(50)
```

Out[28]:
```
array([0.0993562])
```

This treats the continuous variable as continuous, so there is no need to decide an integration width. It is also numerically precise. For error bounds, we can still get them from the trace summary (using the X_2 `mc_error`):

In [29]:
```
pm.summary(mytrace, round_to=5)
```

Out[29]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_mean	ess_sd	ess_bulk
g1	0.89295	0.30918	0.00000	1.00000	0.00462	0.00327	4483.95961	4483.95961	4483.95961
g2	0.81676	0.38687	0.00000	1.00000	0.00544	0.00385	5049.30271	5049.30271	5049.30271
g3	0.99392	0.07774	1.00000	1.00000	0.00036	0.00025	47276.25733	47276.25733	47276.25733
x1	51.05178	4.45846	43.07390	60.54717	0.04947	0.03615	8123.45579	7604.31723	11322.61518
x2	51.81545	5.01304	43.96591	62.72505	0.05797	0.04186	7478.73697	7170.70625	9879.42560

This works because we have a numerically precise value of the probability, so we can port the error estimates over. We couldn't do this using the first window method because we were looking over a range interval, and didn't have a precise estimate when looking at just the trace graphs.

Another thing to note-- because statistics are built of an empirically sampled distribution that is rendered via simulation, we are able to retrieve answers from distributions that are not classic well behaved Gaussian's. The distributions X_1, X_2, and X_3 are not well behaved; they cannot be approximated by a Gaussian, and it's unclear if they could be even be modeled by a combination of classical statical functions. But, using MCMC estimation, we fundamentally don't care what the output distribution looks like; we can query the trace for probability information regardless of it's form. As more explicit example:

In [30]:
```
# We'll complicate our sampling by adding in more data
# pymc3 allows observed variables to be numpy arrays, pandas
# dataframes, or theano data structures; so you can add
# quite a bit of empirical data
```

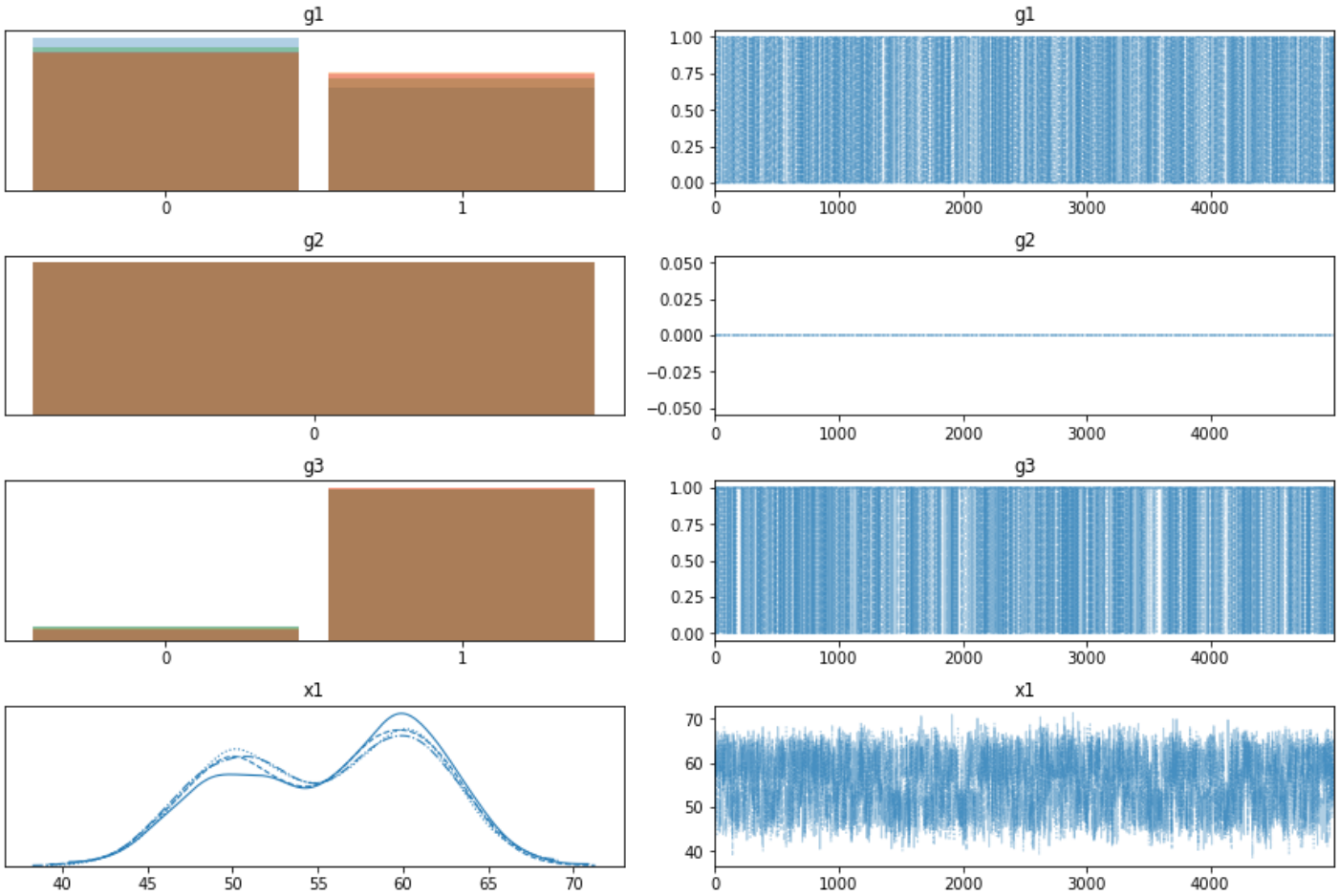
In [31]:
```
data_1 = [38,58,62,55,80,40,51,52]
data_2 = [32,110,72,43,66,55,59,52]
```

In [32]:
```
with Model() as bayesnet:
    # Parent
    g1_healthy = pm.distributions.discrete.Bernoulli('g1',0.5)

    # Latent variable that conditions the children
    g1_L = switch(g1_healthy > 0.5, 0.9, 0.1)

    # Left Child
    g2_healthy = pm.distributions.discrete.Bernoulli('g2', g1_L)
    # Right Child
    g3_healthy = pm.distributions.discrete.Bernoulli('g3', g1_L)
    # Conditional switches for blood pressure mean
    rate1 = switch(g1_healthy > .5, 50, 60)
    rate2 = switch(g2_healthy > .5, 50, 60)
    rate3 = switch(g3_healthy > .5, 50, 60)
    # Grandkids and child
    x1 = Normal('x1', mu=rate1, sd=10**.5)
    x2 = Normal('x2', mu=rate2, sd=10**.5, observed=data_2)
    # Setting x3 as an observed variable
    x3 = Normal('x3', mu=rate3, sd=10**.5, observed=data_1)
```

In [33]:
```
with bayesnet:
    mytrace = pm.sample(5000)
```

```
Multiprocess sampling (4 chains in 4 jobs)
CompoundStep
>BinaryGibbsMetropolis: [g1, g2, g3]
>NUTS: [x1]
Sampling 4 chains, 0 divergences: 100%|██████████| 22000/22000 [01:02<00:00, 350.61draws/
s]
/srv/conda/envs/MLenv/lib/python3.7/site-packages/xarray/core/nputils.py:215: RuntimeWarni
ng: All-NaN slice encountered
  result = getattr(npmodule, name)(values, axis=axis, **kwargs)
The number of effective samples is smaller than 10% for some parameters.
```

In [34]:
```
pm.traceplot(mytrace);
```



Above tells us that the Bayesian estimate is certain that one child has the condition, and is fairly certain that the other doesn't. This means that the parent is more likely to have the condition than not; also note that the blood pressure likelihoods are highest for each case associated with the condition (compared to a mean).